Leakage-controlled benchmarking of radiograph body-part classification: reliability, label audit, explainability and cross-dataset transfer on public data

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Abstract

Background. Published accuracies for radiograph body-part classification are hard to compare: splits, thresholds and uncertainty are rarely reported in detail, and most large studies use private data. **Methods.** We built a leakage-controlled benchmark on the public UNIFESP dataset (22 classes, 1,738 radiographs, 768 patients) with patient-level folds, a 323-image held-out set, all choices made on out-of-fold predictions and patient-bootstrap confidence intervals. We added ablations, a frozen foundation-model baseline, transfer to MURA and FracAtlas, a MURA body-part benchmark, calibration and selective-prediction analysis, and a label audit. **Results.** The ensemble reached a held-out sample-mean F1 of 0.919 (95 % CI 0.867 to 0.960), not significantly above its best single model (+0.0066, $p = 0.063$); no ablation differed from the reference (all $p > 0.08$). A gated CNN-Transformer fusion ignored its CNN branch. A random image-level split overstated validation F1 by 0.0144 (patient-level folds: 0.0004), and tuning thresholds on the scored set added 0.0083. On the official MURA validation set accuracy was 0.9747 (0.9663 to 0.9818); zero-shot transfer gave 0.69 to 0.88. The ensemble was already well calibrated (ECE 0.054), and accuracy on retained images rose from 0.926 to 0.945 at 90 % coverage. The audit flagged 39 of 1,415 training images (2.8 %) as possible label errors, not yet clinician-checked. A late Kaggle submission without pseudo-labels scored 0.9256 (private), between second

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and third place. **Conclusion.** How a classifier is evaluated matters more than its architecture: gains from fusion, ensembling and pseudo-labelling fell within the noise of a 323-image held-out set.

Keywords: body-part classification, radiograph, data leakage, reproducibility, calibration, label noise

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1. Introduction

A radiograph usually reaches the archive with its anatomical region stored in a DICOM tag or a free-text study description, and either can be wrong or missing. An early audit of 562 images found false *Body Part Examined* entries in 15.3 % of them (Güld et al., SPIE 2002). Deep networks can read the region from the pixels instead (Pham et al. 2021; Langer et al. 2024; Mitsuyama et al. 2026). Most of that work uses private data, reports accuracy alone and says little about how the splits and thresholds were chosen. On the public UNIFESP dataset (Farina, Abdala and Kitamura 2022) we found one academic paper, which reports 86.94 % accuracy for the best of five CNNs (Krishnan and Padmavathi 2025). That study treated the 1,738 training images as a single-label 22-class problem, added 1,088 GAN-generated images to the 1,686 originals, and reports accuracy only; we could not find a description of a patient-level split or of a held-out protocol. The same data were also the basis of a Kaggle competition, whose final private leaderboard is led by a score of 0.93378. Two evaluation shortcuts are known to inflate results in other fields: leakage between images of the same patient, and thresholds or checkpoints chosen on data that is later scored (Kapoor and Narayanan 2023; Roberts et al. 2021). We did not find a measurement of how large these effects are for body-part classification. A search of PubMed, arXiv and general web sources in November 2026 found studies of leakage in other imaging tasks but none that measured it for radiograph body-part classification. Fig. 1 shows what the data look like. So instead of chasing the leaderboard we asked two questions. How much of the reported performance survives a leakage-controlled protocol? And what would a clinical router need besides accuracy? The paper contributes:

- a reproducible, patient-level benchmark on public data with confidence intervals (Table 1 and Supplementary Table S1);

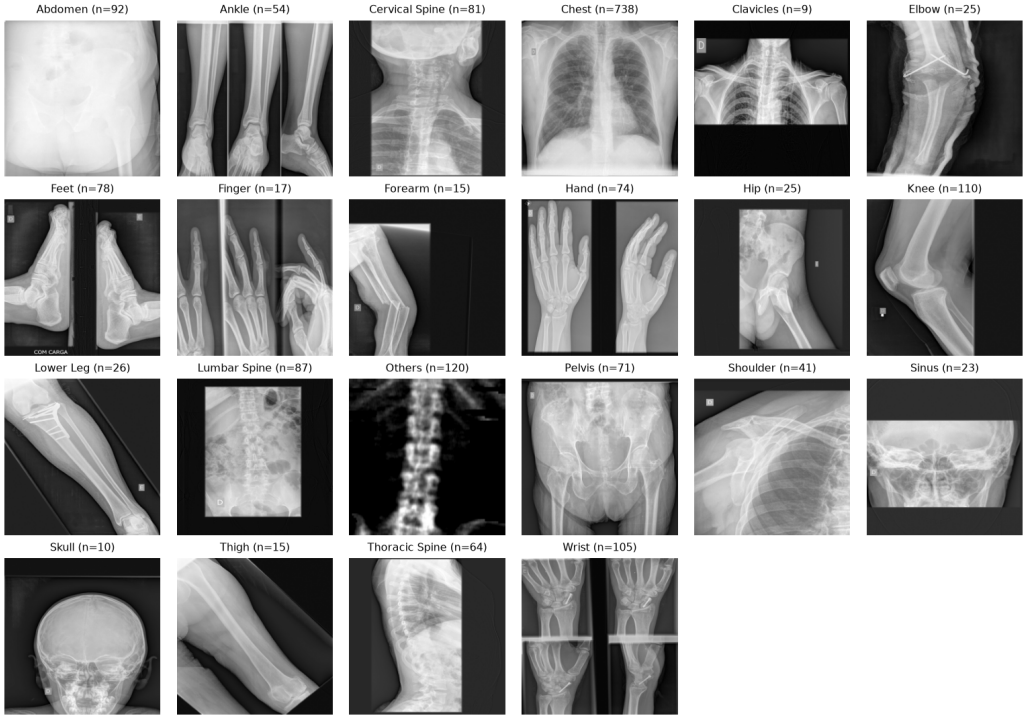


Figure 1: One example radiograph per class from the UNIFESP dataset (CC BY-NC-SA 4.0), with the number of training images in each class. Several images show more than one view side by side.

- a measurement of how much common shortcuts inflate scores (Table 4 and Supplementary Table S2);
- negative results: no ablation, fusion variant or ensembling step was significantly better, and a gated CNN-Transformer fusion collapsed onto one branch (Table 4 and Supplementary Table S3);
- results on MURA and FracAtlas, including a body-part benchmark on the official MURA split (Table 5 to Table 6);
- calibration, risk-coverage and a label audit based on out-of-fold predictions, aimed at routing use (Section 5.7);
- a frozen foundation-model baseline with a label-efficiency curve (Section 5.6) and a Grad-CAM faithfulness check (Supplementary Table S5).

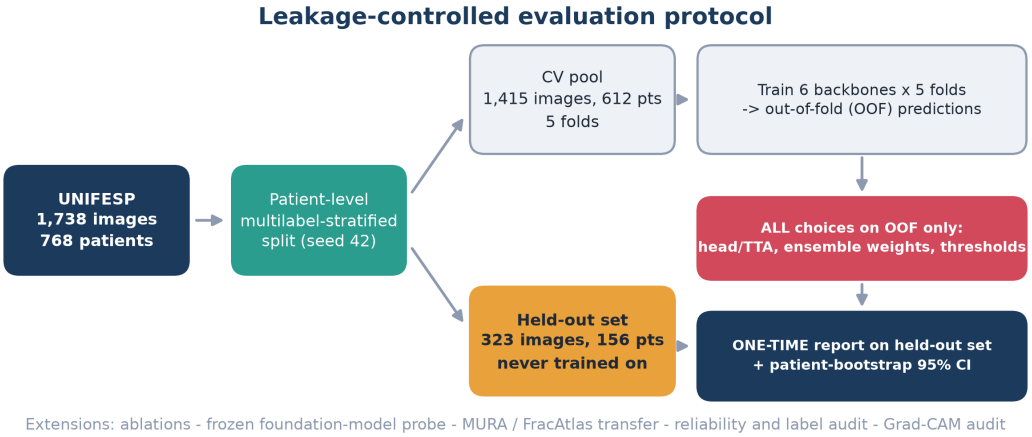


Figure 2: Evaluation protocol. Images are split by patient into a cross-validation pool and a held-out set. Every choice is made on out-of-fold predictions, and the held-out set is scored once, with patient-bootstrap confidence intervals.

Fig. 2 gives the whole protocol at a glance, from the patient-level split to the one-time scoring of the held-out set.

2. Related work

Body-part classification. Pham et al. (2021) sorted 16,093 images into five coarse groups for DICOM routing (recall 0.982) and pointed out that routing needs near-perfect accuracy. Langer et al. (2024, preprint) classified 14 regions on 48,434 hospital radiographs (96.6 % for a single model, 97.7 % for an ensemble). When experts reviewed the test errors, 35 % turned out to be wrong labels and 11 % were out-of-domain images, so label noise accounts for part of the remaining error. Mitsuyama et al. (2026) trained a seven-category model (Head, Neck, Chest, Incomplete Chest, Abdomen, Pelvis, Extremities) on 429,341 radiographs from two institutions and MURA, and tested it on a second institution and on MURA. Every MURA image was labelled "Extremities", and micro-accuracy on MURA was 98.6 % (99 % CI 98.4 to 98.7 %). Radiologist review changed the DICOM-derived body-part label of 3.9 % of the radiographs from one institution and 12.8 % from the other. Their MURA figure cannot be compared with ours: theirs is a coarse seven-way task in which all MURA images share one class, whereas we separate the seven MURA study types. On MURA, Shah et al. (2026) classified the seven upper-extremity study types with an attention-fused Xception-Swin model and reached an accuracy of 0.9643, a macro-F1 of 0.9574 and

an AUC of 0.9963. They used the public validation set (3,197 images) as the test partition, a fixed seed (42), a patient-wise data organisation and image-level evaluation, and they report Wilson confidence intervals for the abnormality-detection accuracies; we did not find intervals or repeated seeds for the body-part result. All of these studies differ from ours in data, class set and metric. **Hybrid CNN-Transformer models** with gated or attention-based fusion are common. We did not find a report that inspects the learned gate, although absence from our search does not prove absence from the literature. **Hierarchical losses** have also been tried: Wu, Tygert and LeCun (2019) found that minimising a hierarchical loss directly is difficult, and that minimising cross-entropy often lowers it almost as well, which makes it more useful as a metric than as a training objective. **Evaluation quality.** Leakage is widespread in machine-learning-based science (Kapoor and Narayanan 2023) and in COVID-19 imaging (Roberts et al. 2021). Label quality limits medical-imaging benchmarks (Oakden-Rayner 2020), and reporting guidance exists in the CLAIM 2024 update (Tejani et al. 2024). Oakden-Rayner (2020) audited MURA labels and found them more accurate than those of ChestX-ray14, although the normal/abnormal labels were unreliable for cases with degenerative joint disease. Confident-learning methods find label errors from out-of-fold probabilities (Northcutt et al. 2021).

3. Data

UNIFESP (Farina, Abdala and Kitamura, Kaggle; CC BY-NC-SA 4.0) holds 2,481 DICOM radiographs. They are de-identified, come from the Universidade Federal de São Paulo under IRB approval, and the Kaggle version is reduced to 256 × 256 pixels. Radiology residents labelled them on the MD.ai platform, with 20 body regions plus "Others", in a multilabel scheme. The dataset page names four residents; it does not say how many readers saw each image, whether disagreements were adjudicated, or who the patients were. There are 1,738 training images from 768 patients (63 % of patients have several images) and 743 test images from 330 patients whose labels are hidden. No patient appears in both sets. Ninety-five percent of the images are MONOCHROME1 (inverted). The DICOM *BodyPartExamined* tag is empty in all 2,481 files, so we could not measure the tag error rate on this dataset. Of the training images, 92.4 % carry one label (122 carry two, 10 carry three), and class frequency runs from 738 (Chest) down to 9 (Clavicles). A random forest on metadata alone (photometric interpretation, bits stored,

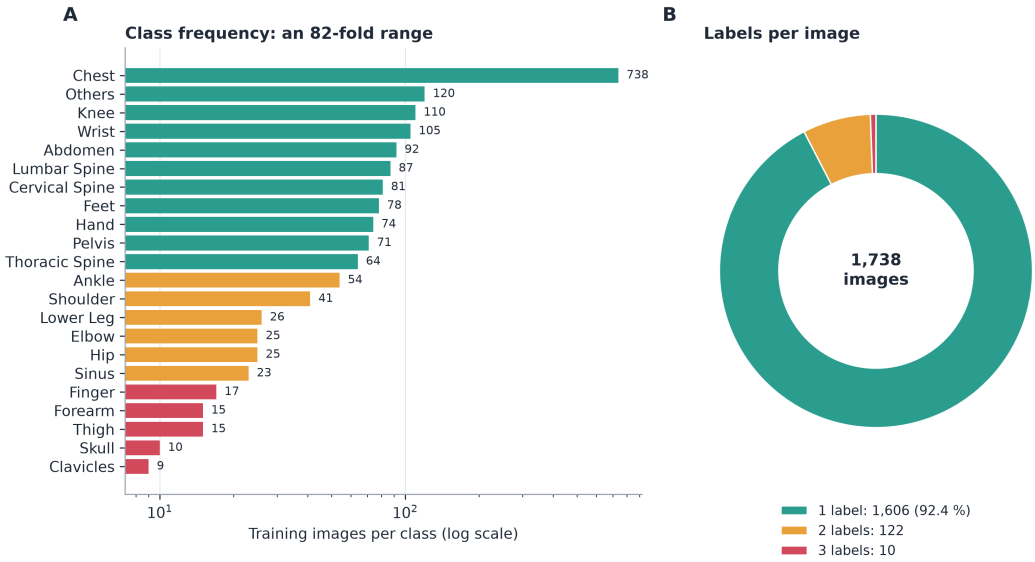


Figure 3: (A) Training images per class, on a log scale. (B) Number of labels per image.

size; patient-grouped cross-validation) reached accuracy 0.486 against 0.451 for the majority class, with macro-F1 0.059, so acquisition metadata hold almost no label information. The class counts are shown in Fig. 3.

Splits. We split by patient with multilabel stratification (iterative stratification, Sechidis et al. 2011; seed 42). That gave a fixed held-out set of 323 images from 156 patients and a cross-validation pool of 1,415 images from 612 patients in five folds. The held-out set has only 1 to 3 positives for Clavicles (1), Skull (2), Forearm (3), Elbow (3) and Thigh (3), so per-class numbers for these classes are unstable. Fig. 4 shows the label combinations (A) and that both parts of the data cover every class.

Other datasets. For **MURA** (Rajpurkar et al. 2018) we took the body part from the `XR_*` folder name (7 classes) and used the official train/validation split. **FracAtlas** (Abedeen et al., Sci Data 2023, CC BY 4.0) has 4,083 images from three hospitals in Bangladesh, annotated by two radiologists and an orthopaedist; we used the 3,685 with a single region (hand, hip, leg or shoulder) for zero-shot transfer. **COVID-19 Radiography** (Chowdhury et al. 2020) has 4 classes and no patient identifiers, so its split is image-level.

Dataset availability. All four datasets are public, but access conditions differ.

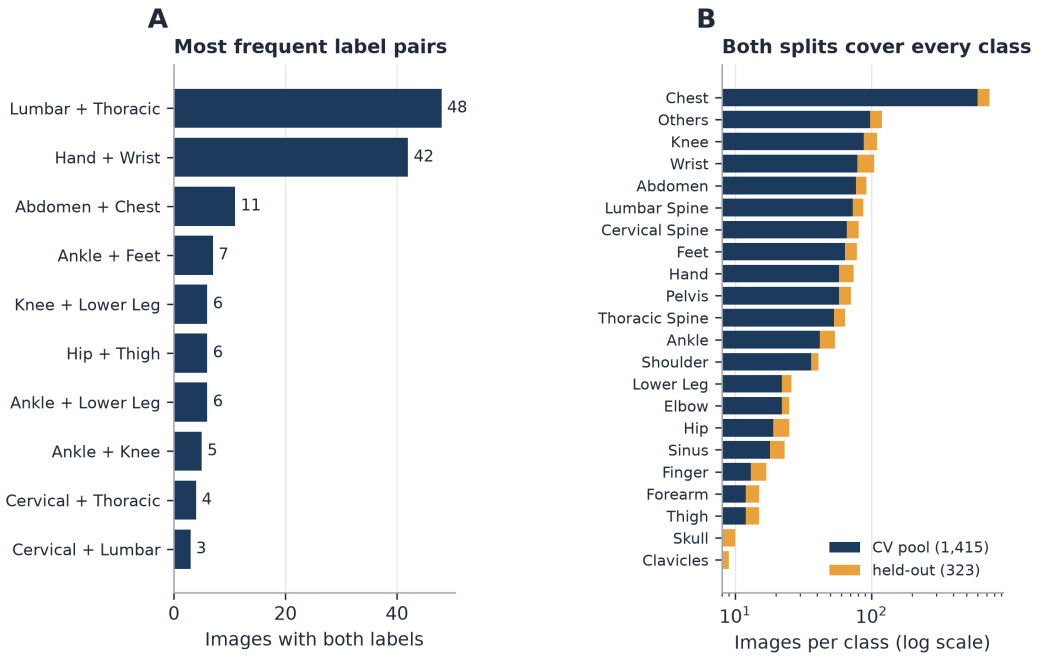


Figure 4: (A) The ten most frequent label pairs; two pairs, Lumbar with Thoracic Spine and Hand with Wrist, account for most multilabel images. (B) Images per class in the cross-validation pool and in the held-out set, on a log scale.

- UNIFESP X-Ray Body Part Classification: `https://www.kaggle.com/datasets/felipekitamura/unifesp-xray-bodypart-classification` (doi:10.34740/KAGGLE/DSV/3399135; CC BY-NC-SA 4.0).

- MURA (v1.0): Stanford AIMI, `https://stanford.redivis.com/datasets/cv1a-apytk3j44` (project page `https://stanfordmlgroup.github.io/competitions/mura/`). Access requires AIMI membership and acceptance of the Stanford University School of Medicine Dataset Research Use Agreement (non-commercial, research-only use; redistribution of the data is not permitted). The first author holds approved access under this agreement (status checked on 6 November 2026). Because redistribution is prohibited we do not release MURA images or derived image files, only code, split lists and result files. The experiments reported here ran on a third-party Kaggle copy of MURA v1.1 (`https://www.kaggle.com/datasets/cjinny/mura-v11`, with no description and no stated licence), which we had downloaded before

the official access was granted. Afterwards we compared all 40,005 image files of that copy with the official Redivis file index. File names, sizes and MD5 hashes (checked on the first 48 bits) were identical, so the copy holds the same images as the official release. The official index also lists four macOS metadata files that are not images. The official validation split served as our held-out set, because the official test set is hidden.

- COVID-19 Radiography Database: `https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database` (the images remain the property of their original authors; see Chowdhury et al. 2020 and Rahman et al. 2021).

- FracAtlas: figshare, doi:10.6084/m9.figshare.22363012 (CC BY 4.0). We used the Kaggle copy `https://www.kaggle.com/datasets/orvile/fracatlas` with 4,083 images as in the Scientific Data paper; the current figshare version (v7) lists 4,073 images, so the two are not identical.

No new patient data were collected. Our code, split files, seeds and result files are public at `https://github.com/NoorAbdullah02/unifesp-bodypart-benchmark` and archived on Zenodo (Abdullah and Hossain 2026; doi:10.5281/zenodo.24027438).

4. Methods

Pre-processing. We applied the DICOM rescale slope and intercept, inverted MONOCHROME1 images, normalised each image to 8 bit by its own minimum and maximum, replicated it to three channels and resized it to 224 pixels (Swin-T) or 256 pixels (all others). ImageNet mean and standard deviation were used for scaling. **Models.** All backbones were ImageNet-pretrained timm models (Wightman 2019): Swin-T (Liu et al. 2021), DenseNet121 (Huang et al. 2017), EfficientNetV2-S (Tan and Le 2021), ConvNeXt-T (Liu et al. 2022), ResNet50 (He et al. 2016) and EfficientNet-B3 (Tan and Le 2019), each with a 22-way sigmoid head and a 22-way softmax head. For the fusion ablation we combined DenseNet121 and Swin-T in five ways: a learned gate on raw features, the same gate with LayerNorm, a fixed 0.5/0.5 average, the CNN branch alone, and the Swin branch alone. Fig. 5 shows the model and the decision pipeline.

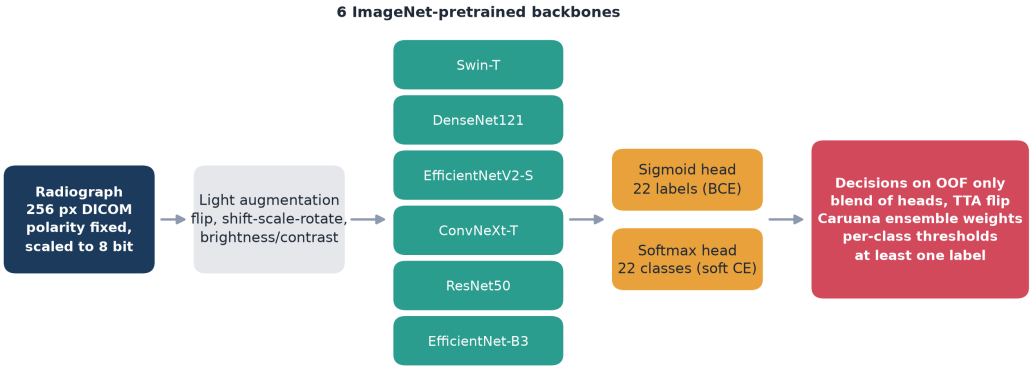


Figure 5: The models and the decision pipeline. Six pretrained backbones, each with a sigmoid and a softmax head, are trained on five patient-level folds. Head blending, ensemble weights, thresholds and the minimum-one-label rule are all chosen on out-of-fold predictions.

Training. We used AdamW (Loshchilov and Hutter 2019; weight decay 1e-4; learning rate 2e-4 for Swin, 3e-4 for DenseNet, ResNet and EfficientNet, 1e-4 for ConvNeXt), a cosine schedule with one warm-up epoch, 30 epochs, batch size 32, fp16, an exponential moving average of the weights (0.998) and gradient clipping at 5. The loss was binary cross-entropy with positive weights capped at 20, plus 0.5 times a soft-target cross-entropy on the softmax head. Light augmentation was flip, shift-scale-rotate and brightness/contrast; heavy augmentation added noise and blur. We did not pick checkpoints: the final averaged weights were used. **Decisions made on out-of-fold (OOF) predictions only.** These were the head and test-time-augmentation variant of each run, the ensemble weights (greedy ensemble selection after Caruana et al. 2004, 40 iterations), the per-class thresholds (coordinate ascent on sample-mean F1, shrunk towards 0.5 with weight n_pos/(n_pos+20)) and the rule that every image gets at least one label. The held-out set was used once (Fig. 2).

Metrics and statistics. We report sample-mean F1 (the competition metric), macro-F1, micro-F1 and subset accuracy. Confidence intervals are 95 % patient-level bootstrap intervals (2,000 resamples, or 1,000 for the reliability analysis), and model comparisons use a paired patient bootstrap. FracAtlas and the other transfer targets have no patient identifiers, so their intervals come from an image-level bootstrap and are optimistic. **Frozen foundation-model baseline.** We froze DINOv2-S and DINOv2-B (Oquab

et al. 2024) and an ImageNet Swin-T at 224 pixels, standardised the features on the cross-validation pool and trained a linear probe with 22 sigmoid outputs (binary cross-entropy plus L2). Weight decay was chosen on OOF predictions, and folds and held-out set were the same as before. To get a label-efficiency curve we repeated this with 10, 25, 50 and 100 % of the training patients in each fold. **Cross-dataset transfer.** We mapped the label spaces onto each other in a closed set (for example MURA hand, wrist and forearm onto the UNIFESP classes of the same name), restricted predictions to the shared groups and report accuracy and macro-F1. The map is our own assumption. **Reliability analysis.** The ensemble combined five saved ablation runs (three Swin-T seeds, DenseNet121 and ConvNeXt-T) with weights fitted on OOF predictions. We measured top-1 calibration (ECE with 15 bins and temperature scaling, Guo et al. 2017; temperature fitted on OOF, applied to the held-out set), drew a risk-coverage curve in which the model abstains on low-confidence images (Geifman and El-Yaniv 2017), and computed per-class precision, recall and F1 with patient bootstrap intervals. For the **label audit**, a training image counts as a *suspect* when the OOF ensemble gives probability 0.9 or more to a class other than its given label and a low probability to the given one (as in confident learning). Suspects are not proven errors. **Explainability.** We computed Grad-CAM maps (Selvaraju et al. 2017) for three fold-0 models on the held-out set, ran a deletion test against a random map on the same grid, and ran a randomisation sanity check (Adebayo et al. 2018). **Stage 2 (competition only).** To see how far more data and pseudo-labels could push the Kaggle score, we repeated the pipeline on all 1,738 training images with no held-out set, so that every image receives an out-of-fold prediction (eight models, called s2a). The s2a ensemble then labelled the 743 unlabelled test images; the 539 predictions with confidence 0.9 or higher were added to every training fold as pseudo-labels (the competition rules allow pseudo-labelling but not manual annotation of the test set), and the same eight models were retrained (s2b). The final entry, s2u, combines all 16 models with weights fitted on the out-of-fold predictions. Stage 2 therefore has no clean held-out set, and it uses test images without their labels; we report it only as a late competition score and as out-of-fold estimates (Section 5.2), not as evidence of generalisation. **Hardware and software.** All runs used Kaggle notebooks on 2 × NVIDIA T4 GPUs (Stage 2 took 18,811 s) with PyTorch 2.11.0 (CUDA 13.0). The other libraries (timm, albumentations, pydicom, scikit-learn) were the versions preinstalled on the Kaggle image at the time of the runs, from October

to November 2026; we did not record a full package list, which we list as a limitation.

5. Results

5.1. Main result (UNIFESP held-out set; Table 1)

The ensemble of six runs, with thresholds tuned on OOF predictions, reached a sample-mean F1 of 0.9189 [0.8669, 0.9602], macro-F1 0.8630 [0.6891, 0.8916], micro-F1 0.9148 and subset accuracy 0.8885. At a plain 0.5 threshold the first two values are 0.9164 and 0.8537. The best single run by OOF (ConvNeXt-T) reached 0.9123, so the ensemble gained +0.0066 [-0.0002, +0.0142] (p = 0.063). Single runs (Supplementary Table S1) lie between 0.910 and 0.922 (Fig. 6).

Table 1: UNIFESP held-out performance (decisions made on out-of-fold predictions only)

decoding	**sample-mean F1**	**macro-F1**	**micro-F1**	**subset acc.**
ensemble, threshold 0.5	0.9164	0.8537	0.9130	0.8885
ensemble, 0.5 + min-one	0.9164	0.8537	0.9130	0.8885
ensemble, per-class thresholds tuned on OOF + min-one	0.9189 [0.8669, 0.9602]	0.8630 [0.6891, 0.8916]	0.9148	0.8885
best single run by OOF (convnext_tiny_s256), 0.5 + min-one	0.9123	0.8482	0.9070	0.8762

Held-out set: 323 images / 156 patients; 95 % CIs by patient-level bootstrap. Ensemble minus best single: +0.0066 [-0.0002, +0.0142], p = 0.063. Ensemble weights (OOF, Caruana): efficientnetv2_s_s256 0.10, convnext_tiny_s256 0.27, resnet50_s256_heavy 0.29, efficientnet_b3_s256_hv 0.34.

5.2. Competition score (Table 2)

Both late submissions are unranked, and we report every submission we made. The Stage 1 ensemble scored 0.92563 on the private leaderboard and 0.92704 on the public one; the Stage 2 ensemble of 16 models scored 0.92638 and 0.93378. The final private leaderboard has 0.93378 in first place, 0.92787 in second and 0.92488 in third, so both of our submissions would sit between second and third place.

The private score uses about 60 % of the 743 test images and their original, partly noisy labels. We estimate its standard error at about 0.012, which means the gap to first place (0.0074) cannot be resolved and the gain of Stage 2 over Stage 1 (+0.00075) is indistinguishable from noise. Stage 2 was evaluated differently, with OOF predictions on all 1,738 training images, so its numbers are not comparable with the held-out results (Table 3). The

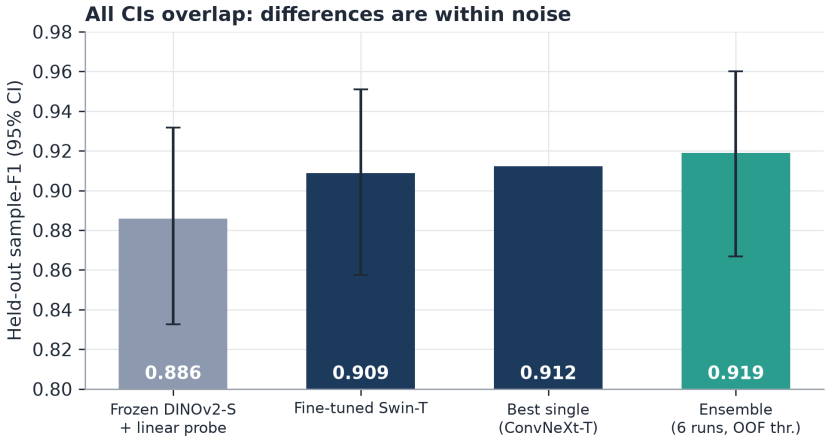


Figure 6: Held-out sample-F1 for a frozen DINOv2-S probe, the fine-tuned Swin-T, the best single model and the ensemble. All 95 % confidence intervals overlap.

Table 2: Kaggle UNIFESP private score vs. the competition leaderboard

entry	**private score**	**public score**
#1 Harshit Sheoran	0.93378	
#2 JGK (ResNet50/152, FastAI)	0.92787	
ours: stage-2 union (late submission)	0.92638	0.93378
ours: stage-1 ensemble (late submission)	0.92563	0.92704
#3 Colin Greeley	0.92488	
#4 KobeU Systems Medicine	0.92414	
#5 wangkui	0.92182	

Mean F1 (sample-wise); private score uses ˜60 % of the test set with the original (partly noisy) labels, so differences of ˜0.003 are within noise. Late submissions are not listed on the leaderboard; we report every submission we made.

plain ensemble of 8 models scored 0.9262 [0.9092, 0.9414]. The same models retrained with pseudo-labels from 539 of the 743 unlabelled test images (confidence 0.9 or higher) scored 0.9233 [0.9058, 0.9392], and the union of all 16 models scored 0.9279 [0.9116, 0.9428]. In paired patient-bootstrap comparisons the union beat the plain ensemble by +0.0017 [-0.0010, +0.0046] ($p = 0.25$), and the pseudo-labelled models fell short of it by 0.0029 [-0.0092, +0.0034] ($p = 0.36$). Pseudo-labelling on its own did not help. It lowered both OOF F1 and macro-F1 (the latter from 0.8749 to 0.8587). The union's small gain rests on weights fitted to the same OOF predictions, so it is op-

timistic. The pseudo-labelled set contains no Hand, Hip, Lumbar Spine, Thoracic Spine, Skull, Thigh, Forearm or Clavicles images and 322 Chest images, so it leans heavily towards frequent classes. Because the procedure uses test images, we do not report it as evidence of generalisation.

Table 3: Stage 2: out-of-fold results on all 1,738 training images and Kaggle late-submission scores

ensemble	OOF sample-F1 [95% CI]	macro-F1	subset acc.	Kaggle private / public
s2a: 8 models, no pseudo-labels	0.9262 [0.9092, 0.9414]	0.8749	0.8867	not submitted
s2b: 8 models, pseudo-labels	0.9233 [0.9058, 0.9392]	0.8587	0.8855	not submitted
s2u: union of s2a and s2b (16 models)	0.9279 [0.9116, 0.9428]	0.8754	0.8884	0.92638 / 0.93378
difference s2u - s2a	+0.0017 [-0.0010, +0.0046], p = 0.25			
difference s2b - s2a	-0.0029 [-0.0092, +0.0034], p = 0.36			

OOF numbers are not comparable with the held-out numbers of Table 1 (different protocol: no fixed held-out set). Ensemble weights are fitted on the same OOF predictions, so the union estimate is optimistic. CIs: patient-level bootstrap (2,000 resamples); pseudo-labels use unlabelled test images (transductive).

5.3. Evaluation shortcuts (Table 4 and Supplementary Table S2)

With a random image-level split, the validation score exceeded the held-out score by 0.0144; with patient-level folds the difference was 0.0004. The held-out score of the leaky model (0.9163) is not significantly different from the reference ($p = 0.141$), so leakage distorted the validation estimate rather than the held-out result. Tuning per-class thresholds on the held-out set itself gave 0.9182, against 0.9099 when the thresholds were tuned on OOF predictions (+0.0083). That difference is a gain taken from the test set. Our earlier notebooks, which tuned thresholds and chose checkpoints on the scored fold, reported 0.9356 (sample-F1) and 0.909 (macro-F1). Under the nested protocol the same quantities are 0.9189 and 0.8630 (Fig. 7A).

5.4. Ablations (Table 4)

Against the reference Swin-T (0.9089 [0.8576, 0.9509]) no configuration differed significantly. Heavy augmentation changed F1 by -0.0075 ($p = 0.088$), dropping the softmax head by -0.0051 ($p = 0.152$), two other seeds by +0.0055 and +0.0065 ($p = 0.37$ and 0.27), DenseNet121 by +0.0111 ($p = 0.119$), the CNN-only fusion branch by +0.0117 ($p = 0.097$) and the average of three seeds by +0.0049 ($p = 0.38$). All 95 % intervals include zero. The seed-to-seed spread (0.9089 to 0.9154) is as large as most of these effects, so a single-run difference of this size should not be called an improvement (Fig. 8).

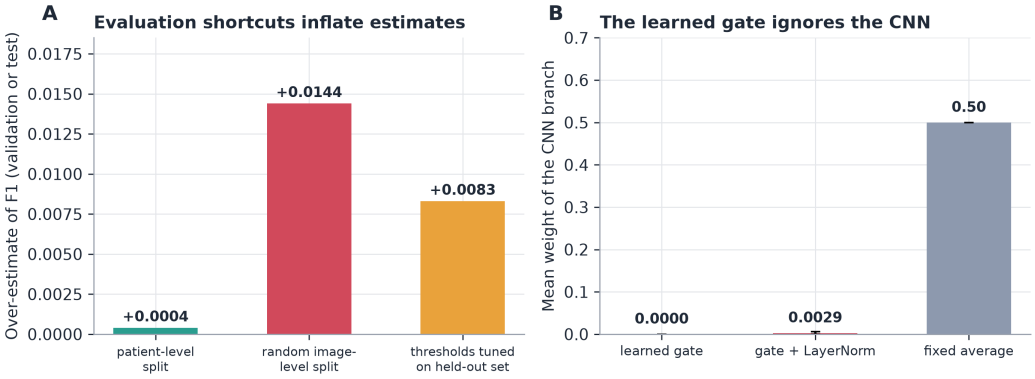


Figure 7: (A) How much a random image-level split and tuning thresholds on the held-out set overstate the score. (B) Mean weight the learned fusion gates give the CNN branch, compared with a fixed average.

Table 4: Ablations (5 CV folds each, same held-out set; paired patient bootstrap vs. the reference)

config	held-out sample-F1 [95% CI]	macro-F1	OOF - held-out	diff vs reference [95% CI]	p
swin_tiny_s224	0.9089 [0.8576, 0.9509]	0.837	0.0004		
swin_tiny_s224_heavy	0.9013 [0.8494, 0.9440]	0.829	0.0074	-0.0075 [-0.0164, 0.0009]	0.088
swin_tiny_s224_ce0	0.9038 [0.8529, 0.9470]	0.8261	0.0041	-0.0051 [-0.0124, 0.0016]	0.152
swin_tiny_s224_ce1	0.9132 [0.8616, 0.9568]	0.8495	0.0012	0.0043 [-0.0063, 0.0149]	0.447
swin_tiny_s224_leaky	0.9163 [0.8644, 0.9587]	0.856	0.0144	0.0074 [-0.0020, 0.0178]	0.141
swin_tiny_s224_seed1	0.9143 [0.8620, 0.9576]	0.8533	0.0028	0.0055 [-0.0060, 0.0173]	0.366
swin_tiny_s224_seed2	0.9154 [0.8635, 0.9592]	0.8585	-0.0014	0.0065 [-0.0049, 0.0178]	0.268
hybrid_gate_s224	0.9136 [0.8609, 0.9568]	0.8497	0.0000	0.0047 [-0.0063, 0.0161]	0.401
hybrid_gate_ln_s224	0.9158 [0.8641, 0.9577]	0.8536	-0.0019	0.0069 [-0.0032, 0.0182]	0.226
hybrid_avg_s224	0.9164 [0.8651, 0.9585]	0.8537	-0.0004	0.0075 [-0.0048, 0.0201]	0.235
hybrid_cnn_s224	0.9205 [0.8684, 0.9613]	0.8705	-0.0012	0.0117 [-0.0016, 0.0257]	0.097
hybrid_swin_s224	0.9127 [0.8611, 0.9565]	0.8467	0.0013	0.0038 [-0.0070, 0.0145]	0.539
densenet121_s256	0.9200 [0.8694, 0.9618]	0.8542	-0.0042	0.0111 [-0.0022, 0.0275]	0.119
convnext_tiny_s256	0.9117 [0.8605, 0.9546]	0.8472	0.0081	0.0028 [-0.0084, 0.0142]	0.674
swin_3seeds_average	0.9137 [0.8623, 0.9574]	0.848	0.0012	0.0049 [-0.0052, 0.0148]	0.38

`OOF - held-out` large and positive = the validation score over-estimates (compare `_leaky` with the patient-level reference for the split ablation).

5.5. Fusion gate (Supplementary Table S3)

Across five folds the learned gate gave the CNN branch a mean weight of 0.0000 (raw gate) and 0.0029 (LayerNorm gate). The fused model therefore behaves like the Transformer alone (Fig. 7B). Neither gate beat the fixed average (+0.0047 and +0.0069 against the reference, p = 0.40 and 0.23) or the CNN-only branch.

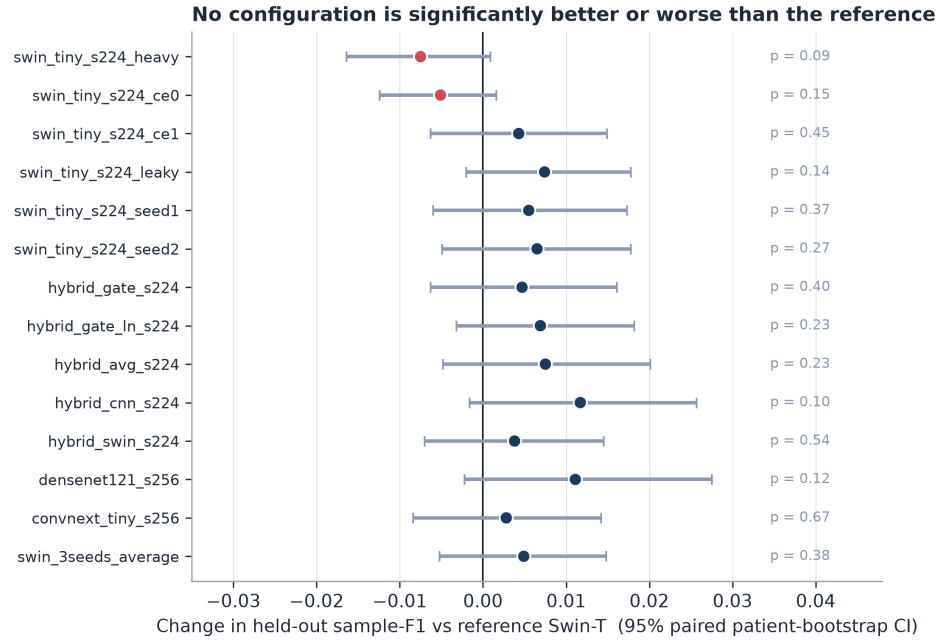


Figure 8: Change in held-out sample-F1 against the reference Swin-T for each ablation (95 % paired patient-bootstrap interval; p-values on the right). None of the intervals excludes zero.

5.6. Frozen foundation-model baseline

A linear probe on frozen DINOv2-S reached 0.8699 OOF and 0.8859 on the held-out set [0.8327, 0.9318]. DINOv2-B reached 0.8839 [0.8300, 0.9276] and the ImageNet Swin-T 0.8762 [0.8248, 0.9204], while the fine-tuned Swin-T reference reached 0.9089. Frozen features therefore recover most of the fine-tuned performance, but not all of it. With 10 % of the training patients the DINOv2-S probe scored 0.6435 OOF, with 25 % 0.7457, with 50 % 0.8235 and with all of them 0.8699. The curve is still rising, so this task looks limited by labelled data. A random-weight ResNet18 control gave 0.563 OOF but 0.095 on the held-out set, an inconsistency we could not resolve, so we leave it out of our claims. BiomedCLIP could not be downloaded and one ConvNeXt tag was invalid; both were skipped (Fig. 9).

5.7. Reliability and label audit

Calibration. Top-1 accuracy of the five-model ensemble on the held-out set was 0.9257. ECE was 0.0545, and 0.0517 after temperature scaling ($T = 1.15$), so the ensemble is already close to calibrated and scaling changes little.

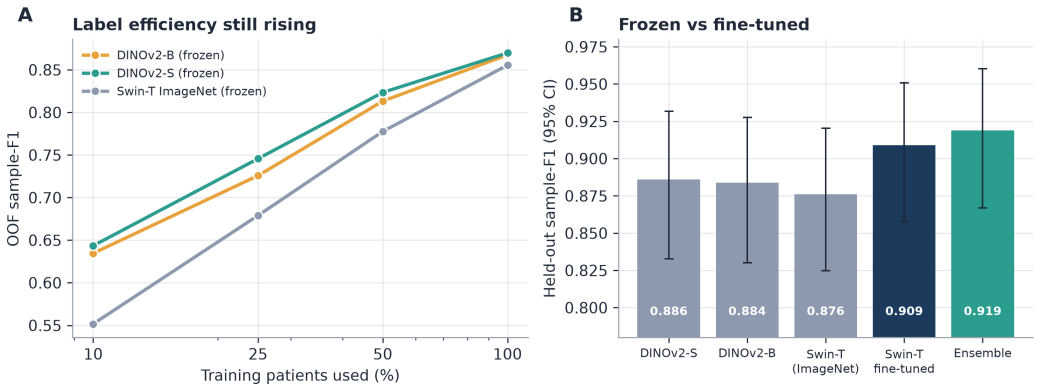


Figure 9: (A) Label efficiency of frozen encoders. (B) Held-out F1 of the frozen probes next to the fine-tuned models.

Selective prediction. Accuracy on the images the model kept rose from 0.926 at full coverage to 0.945 [0.900, 0.985] at 90 % coverage, 0.973 at 80 % and 0.982 at 70 %. A threshold chosen on OOF predictions for 0.95 accuracy gave 0.937 accuracy at 98.5 % coverage on the held-out set, so the target was missed. A 0.98 target gave 0.983 at 74 % coverage. Operating points chosen on OOF predictions therefore do not transfer exactly (Fig. 10).

Per class (held-out). Chest reached an F1 of 0.993 (n = 138), and Cervical Spine, Feet, Hand, Sinus, Skull and Thigh reached 1.0. The weakest classes were Others (recall 0.41 [0.13, 0.91]), Forearm (0.33), Hip (0.33), Shoulder (0.60) and Ankle (0.75), all with 3 to 22 positives and wide intervals (Supplementary Table S7 and Supplementary Fig. S1).

Label audit. At p of 0.9 or more, 39 of 1,415 training images (2.8 %) were flagged. The most frequent pairs were Hip to Pelvis (5), Wrist to Ankle (4) and Others to Chest (3). The Hip-to-Pelvis pattern mirrors the low Hip recall, which fits either label noise or a real boundary problem between the two classes. No clinician has reviewed any of the 39 suspects.

Error analysis (Supplementary Table S6). In the three explained fold-0 models the most frequent held-out misses were Others labelled as Abdomen (6 to 8 images per model) and Others labelled as Pelvis (4 to 5), followed by Hip as Pelvis (2 per model) and Forearm as Lower Leg (2 per model). The heterogeneous "Others" class is mostly absorbed by large body regions. Fig. 11 shows which label pairs the suspects fall into (A) and how the three explained models misclassify the held-out images (B).

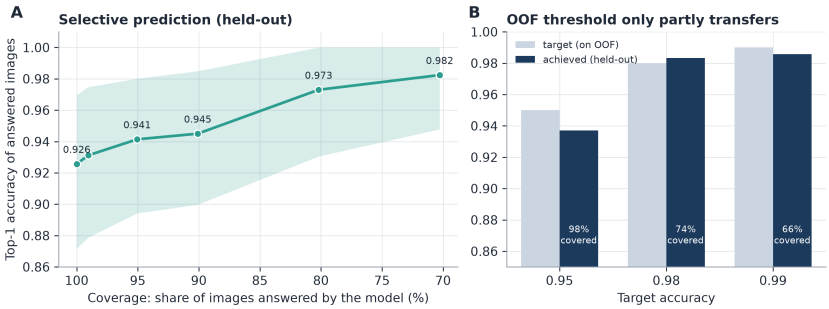


Figure 10: (A) Accuracy on the images the model answers, as a function of coverage. (B) Accuracy targets chosen on out-of-fold predictions, and the accuracy reached on the held-out set.

5.8. Generalisation (Table 5 and Table 6; Supplementary Table S4)

On the official MURA split the five-fold bagged ensemble reached an accuracy of 0.9747 [0.9663, 0.9818] and a macro-F1 of 0.9707 [0.9616, 0.9788]. Shah et al. (2026) report 0.9643 and 0.9574 on the same 3,197-image validation set, so the two results are comparable in data but not in method details. On COVID-19 Radiography (4 classes, image-level split) sample-F1 was 0.9647 [0.9595, 0.9697] and macro-F1 0.9681, and these numbers may be optimistic. Zero-shot transfer (Table 6) gave 0.875 to 0.880 from MURA to UNIFESP (n = 184) and 0.813 to 0.821 from UNIFESP to MURA (n = 12,000). From MURA to FracAtlas, restricted to hand and shoulder (1,281 of the 1,382 images are hand), accuracy was 0.691 with the softmax head and 0.858 with the sigmoid head. From UNIFESP to FracAtlas (hand, hip, leg, shoulder; n = 3,685) it was 0.708 and 0.698. Many FracAtlas JPEGs are truncated and were decoded only partly. Fig. 12 summarises the MURA comparison (A) and the transfer results (B).

Transfer is moderate and depends on the head, and the class maps are our own assumptions.

5.9. Explainability (Supplementary Table S5)

For all three models, deleting the most salient pixels first removed class evidence faster than deleting a random block of the same size (gain 0.284 [0.252, 0.316] for ConvNeXt-T, 0.204 for ResNet50, 0.164 for Swin-T). The maps were uncorrelated with those of randomly initialised networks for ConvNeXt-T and Swin-T (rho 0.11 and 0.02). ResNet50 differed too, but with a systematic negative correlation (rho -0.31). The feature grids are coarse (7×7 to

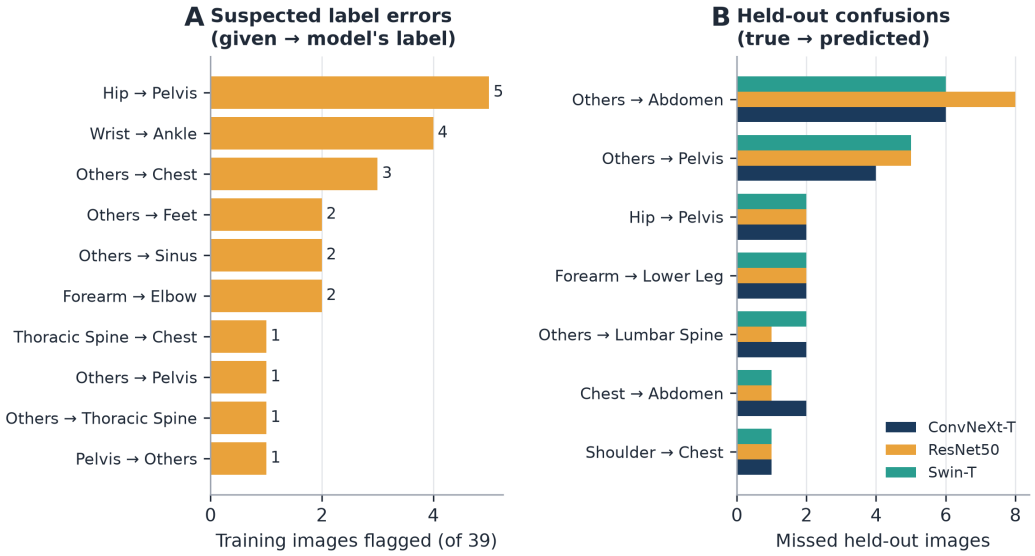


Figure 11: (A) The pairs of given and suggested labels among the 39 suspected label errors. (B) The most frequent misclassifications of the three explained models on the held-out set.

Table 5: MURA body-part classification: held-out = MURA validation set

model	**accuracy [95% CI]**	**macro-F1 [95% CI]**
ours: 5-fold bagged ensemble (official MURA split)	0.9747 [0.9663, 0.9818]	0.9707 [0.9616, 0.9788]
Xception-Swin (Shah 2026)	0.9643 (CI n/a)	0.9574 (CI n/a)
DenseNet-201	0.9187 (CI n/a)	0.9093 (CI n/a)
EfficientNet-B0	0.9043 (CI n/a)	0.8952 (CI n/a)
ResNet-101	0.8934 (CI n/a)	0.8831 (CI n/a)
VGG-19	0.8612 (CI n/a)	0.8498 (CI n/a)

Shah et al. 2026 (Supplementary Table S4; verified in the full text): same MURA validation set (3,197 images), fixed seed 42, image-level evaluation; no CI is shown for these body-part numbers.

8×8), there is no localisation ground truth and no radiologist reviewed the maps. We therefore describe them as faithful to the model, not as clinically valid (Fig. 13).

6. Discussion

Evaluation mattered more than architecture here. Ensembling, fusion, heavy augmentation, extra seeds and pseudo-labelling each moved the held-out score by about 0.01 at most, none of them significantly, and seed noise was

Table 6: Zero-shot cross-dataset transfer through the class map

source	target	head	groups	n	accuracy	95 % CI	macro-F1
mura	unifesp	softmax	elbow,finger,forearm,hand,shoulder,wrist	184	0.875	[0.826, 0.924]	0.8478
mura	unifesp	sigmoid	elbow,finger,forearm,hand,shoulder,wrist	184	0.8804	[0.837, 0.924]	0.856
unifesp	mura	softmax	elbow,finger,forearm,hand,shoulder,wrist	12000	0.8133	[0.806, 0.820]	0.7444
unifesp	mura	sigmoid	elbow,finger,forearm,hand,shoulder,wrist	12000	0.8208	[0.814, 0.828]	0.7481
mura	fracatlas	softmax	hand,shoulder	1382	0.691	[0.667, 0.715]	0.5606
mura	fracatlas	sigmoid	hand,shoulder	1382	0.8575	[0.840, 0.875]	0.7115
unifesp	fracatlas	softmax	hand,hip,leg,shoulder	3685	0.708	[0.693, 0.722]	0.7409
unifesp	fracatlas	sigmoid	hand,hip,leg,shoulder	3685	0.6982	[0.684, 0.713]	0.699

Closed-set decision among the shared groups; CIs are image-level (the targets have no patient ids). The class map is an assumption to be checked by a clinician.

about as large. Papers that report differences of this size without intervals deserve some caution. Most errors sit in the rare classes (1 to 6 held-out positives) and in the heterogeneous "Others" class. The Hip-to-Pelvis cluster in the audit, and the competition hosts' offline re-annotation of the test set, suggest that label noise sets part of the ceiling, as Langer et al. (2024) found on a hospital dataset. The label-efficiency curve was still rising, so more labelled data would probably help more than a different architecture. We did not test why the gate collapsed. One possibility is that a gate learned on the raw features of two networks with different activation scales lets one branch dominate. The learned weights point to a degenerate solution, not to complementary fusion, and fixed averaging was no worse. For clinical use, a router that answers 90 % of the images keeps 94.5 % top-1 accuracy and sends the other 10 % to a person. The operating point chosen on OOF predictions should be re-validated on local data before anyone relies on it. We see the classifier as a way to flag suspicious metadata, not as a replacement for it. Our MURA accuracy is higher than the one published figure we found for this task. Other studies report 97 to 99 % on other data and with much larger private training sets (Langer et al. 2024; Mitsuyama et al. 2026), and we do not claim to outperform them.

7. Limitations

The dataset comes from a single centre and has 1,738 labelled images, and the held-out set of 323 images has only 1 to 6 positives for several classes. The COVID and FracAtlas results use image-level splits and intervals. The class maps used for transfer are ours and nobody has reviewed them, and neither the label audit nor the Grad-CAM maps have been checked by a

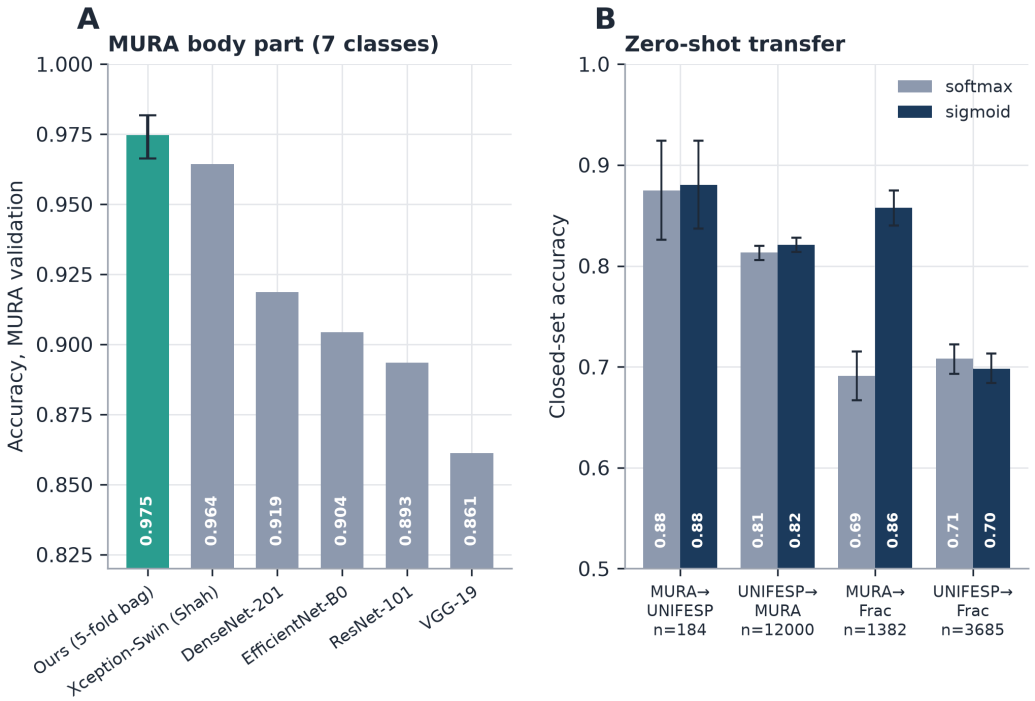


Figure 12: (A) Body-part accuracy on MURA (seven classes), compared with the models reported by Shah et al. (2026). (B) Zero-shot accuracy through the class map, for the softmax and sigmoid heads; intervals come from an image-level bootstrap and are optimistic.

clinician. The private test labels are noisy. Stage 2 uses unlabelled test images (a transductive step) and weights fitted on OOF predictions. The explainability analysis covers one fold per model. The results come from several Kaggle accounts, although all of them can be reproduced from the released code. One control (the random-weight probe) was inconsistent and is excluded. We did not record a full package list for the Kaggle runs, and we had no external clinical site.

8. Conclusion

Under a leakage-controlled protocol an ensemble reaches a sample-F1 of 0.919 (0.867 to 0.960) on public UNIFESP data. Fusion, extra seeds and pseudo-labels add nothing we can tell apart from noise, while random splits and test-set thresholds inflate the estimates. On the official MURA validation set body-part accuracy is about 97 %. The ensemble is reasonably calibrated

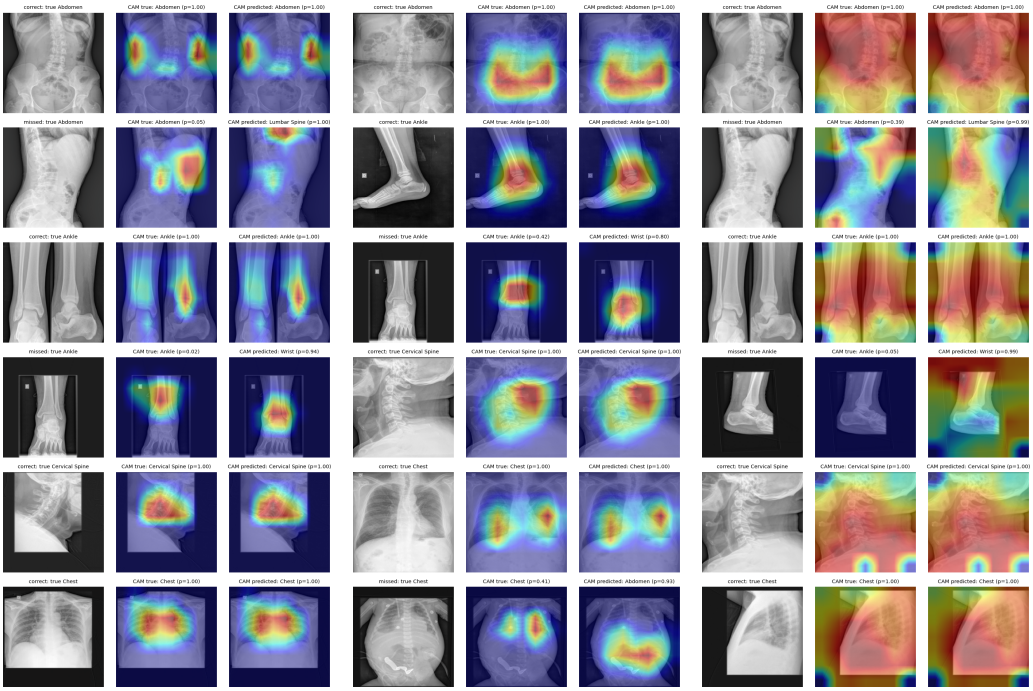


Figure 13: Grad-CAM galleries for ConvNeXt-T (left), ResNet50 (middle) and Swin-T (right). For each class there is one correct and one missed held-out image, with the map for the true class and for the predicted class.

and can abstain on uncertain images, and the label audit hands experts a short list of 39 images to check.

9. Declarations

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10. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used Claude (Anthropic) for code and drafting support, then reviewed and edited everything and take full responsibility for the content.

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